

Lymphoid Tissue Histopathology

No dedicated Word companion • built from official LOs + Week 7 lymphoid summaries

Official lab LOs

No one-to-one doc

Normal-organ recognition

Reactive lymphadenopathy framework

2 ScholarRx Bricks

PRE-SAL VERSION. There is no dedicated Lymphoid Tissue Histopathology Word companion. This draft uses the official lab objectives and the normal lymphoid histology already taught in Hematopoiesis & the Lymphoid System. The reactive-lymphadenopathy section is a provisional pathology framework to help before class; it should be verified against the instructor's actual SAL images and emphasis afterward.

Official Lab Objectives

1. Recognize normal histology of cells, tissues, and structures of lymph nodes, tonsil, gut-associated lymphoid tissue, spleen, and thymus.
2. Recognize characteristic pathological changes in lymph nodes as a result of lymphadenopathy.

SCHOLARRX ASSIGNED

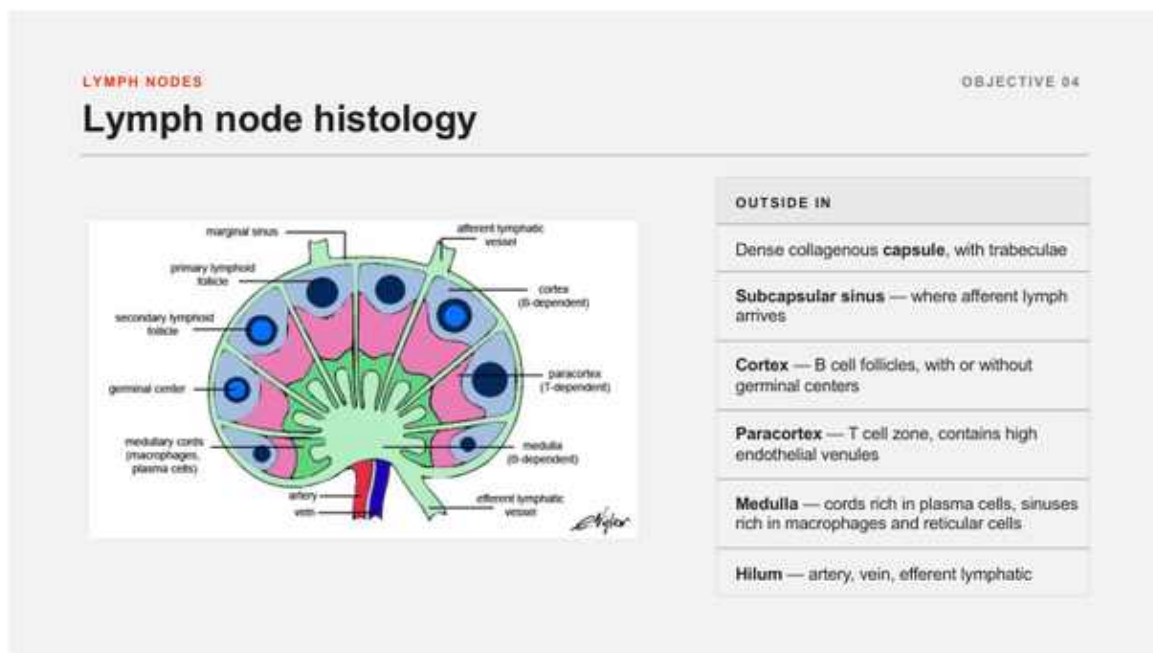
Histology & Function of Lymph Nodes

Benign Lymph Node Disorders & Lymphadenopathy

THE WHOLE PRACTICAL IN ONE QUESTION

What organ is this, and is its normal architecture preserved, reactively expanded, or replaced/effaced?

Lymph Node



Cross-reference from Week 7 Hematopoiesis & Lymphoid System, not an SAL slide. Cross-reference from Lecture 3. Learn the outside-in map now; replace/add the actual SAL specimen later.

OUTSIDE -> IN

Capsule -> subcapsular sinus -> cortex (B-cell follicles) -> paracortex (T cells + HEVs) -> medulla (cords + sinuses) -> hilum.

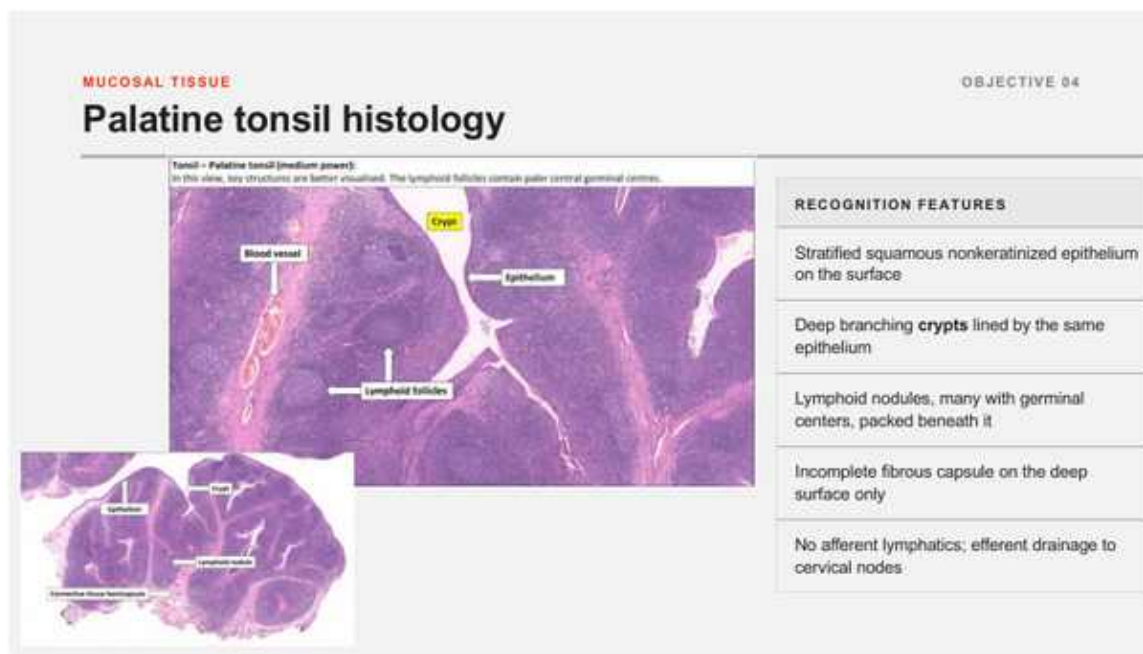
Zone	Normal histology	Key function
Cortex	B-cell follicles. Primary follicles are dark/uniform; secondary follicles have pale germinal centers.	B-cell activation and germinal-center reactions.
Paracortex	T-cell-rich zone between follicles and medulla; contains high endothelial venules.	Naive lymphocytes enter from blood through HEVs; T-cell/APC interactions.
Medullary cords	Cellular cords rich in plasma cells, lymphocytes, macrophages.	Antibody-producing plasma-cell compartment.
Medullary sinuses	Paler channels containing macrophages/reticular cells.	Filter lymph on its way to the hilum.

PRIMARY VS SECONDARY FOLLICLE

Primary follicle: small/dark resting B cells, no pale center. **Secondary follicle:** pale germinal center + darker mantle zone, meaning active antigen-driven B-cell proliferation.

SAL IMAGE SLOT: normal node low power • primary follicle • secondary follicle/germinal center • HEV • medullary cords/sinuses

Tonsil & Gut-Associated Lymphoid Tissue



Cross-reference from Week 7 Hematopoiesis & Lymphoid System, not an SAL slide. Cross-reference from Lecture 3: nonkeratinized stratified squamous epithelium + deep crypts + underlying lymphoid follicles.

Structure	Recognition clue
Palatine tonsil	Nonkeratinized stratified squamous epithelium, deep branching crypts , many lymphoid nodules, incomplete deep capsule.
Pharyngeal tonsil / adenoid	Respiratory epithelium; folds rather than the deep squamous crypts of palatine tonsil.
Peyer's patch	Ileum ; aggregated lymphoid follicles in mucosa/submucosa, dome/M-cell region, no true capsule.
Appendix	Narrow irregular lumen, colonic crypts, no villi , circumferential lymphoid follicles.

Peyer's patch and gut nodules



RECOGNITION FEATURES

Aggregated follicles in mucosa and submucosa of the ileum

Antimesenteric wall, unencapsulated

Primary nodule — no germinal center; **secondary nodule** — pale germinal center

Dome region with villi absent over the follicle

M cells in the follicle-associated epithelium transcytose luminal antigen

Output: dimeric IgA released into the lumen with secretory component, which resists proteolysis.

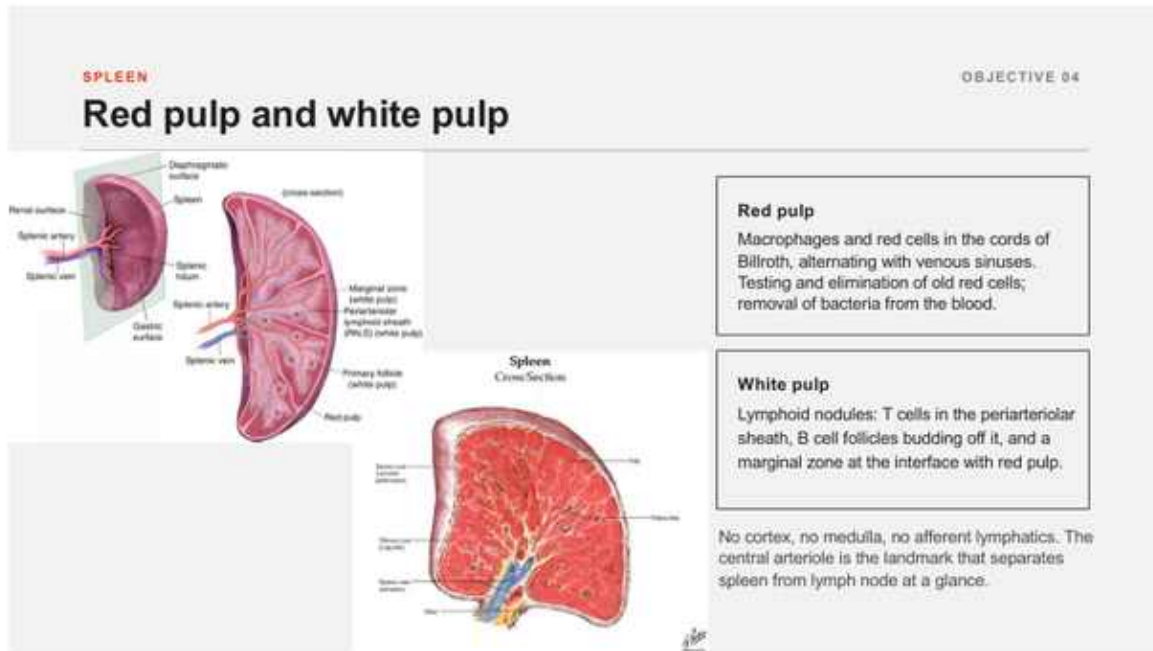
Cross-reference from Week 7 Hematopoiesis & Lymphoid System, not an SAL slide. Cross-reference from Lecture 3. Aggregated ileal follicles and overlying antigen-sampling epithelium are the visual anchor.

MUCOSAL RULE

Lymphoid tissue sits directly under epithelium and is generally unencapsulated; antigen can cross the epithelium directly.

SAL IMAGE SLOT: palatine vs pharyngeal tonsil • Peyer's patch • appendix

Spleen



Cross-reference from Week 7 Hematopoiesis & Lymphoid System, not an SAL slide. Cross-reference from Lecture 3. The splenic signature is red pulp + white pulp centered around arterioles, not cortex/paracortex/medulla.

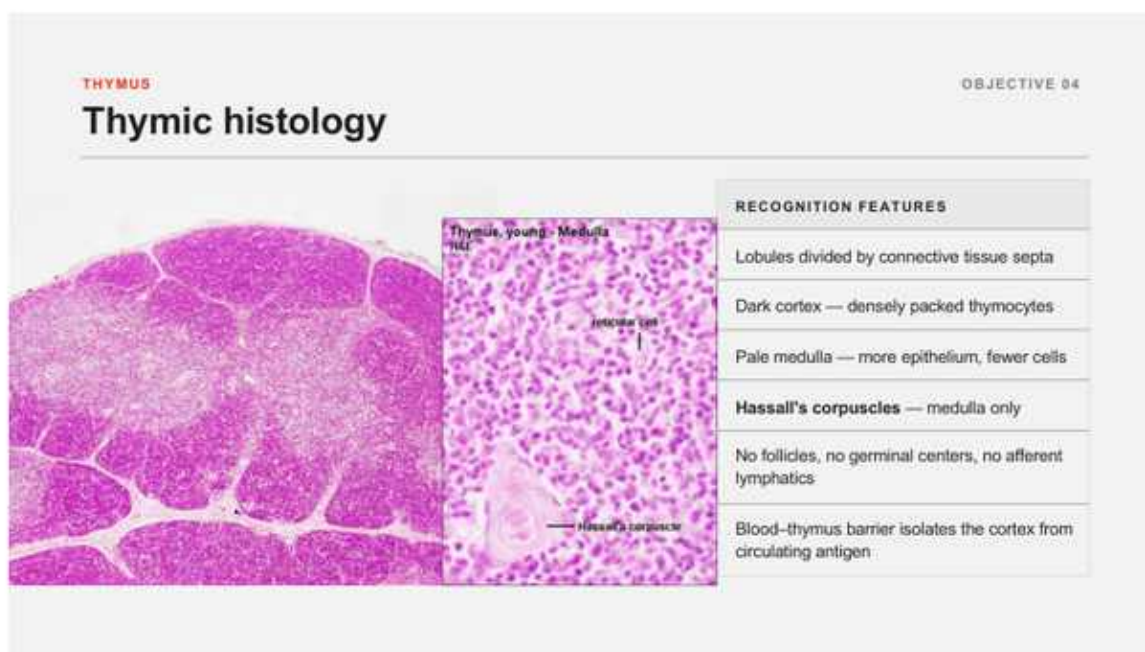
Region	Recognition clue
White pulp - PALS	T-cell sheath around a central arteriole.
White-pulp follicle	B-cell nodule adjacent to PALS; may have a germinal center when active.
Marginal zone	Interface between white and red pulp; macrophages + memory B cells.
Red pulp	RBC-rich cords of Billroth and venous sinusoids; mechanical blood filtration.

LYMPH NODE VS SPLEEN

Lymph node: capsule + cortex/paracortex/medulla + subcapsular sinus. **Spleen:** red/white pulp + central arteriole, no cortex/medulla and no afferent lymphatics.

SAL IMAGE SLOT: spleen low power • PALS/central arteriole • red pulp/sinus

Thymus



Cross-reference from Week 7 Hematopoiesis & Lymphoid System, not an SAL slide. Cross-reference from Lecture 3: lobules, dark cortex, pale medulla, and Hassall corpuscles.

THYMUS COLD SET

Lobulated organ + dark cortex + pale medulla + Hassall corpuscles = thymus.
No follicles. No germinal centers. No afferent lymphatics.

Region	What you see / function
Cortex	Very dark because packed with immature thymocytes; positive selection occurs here.
Medulla	Paler, fewer lymphocytes, more epithelium; negative selection and Hassall corpuscles.
Hassall corpuscle	Concentric eosinophilic epithelial whorl in medulla only.

SAL IMAGE SLOT: thymus low power • cortex/medulla • Hassall corpuscle

Reactive Lymphadenopathy - Provisional Pre-SAL Framework

IMPORTANT

The official objective says to recognize pathological lymph-node changes caused by lymphadenopathy, but there is **no dedicated summary or released SAL image set yet**. The patterns below are the standard benign/reactive framework to study provisionally. We will revise this section to match exactly what your SAL instructor shows.

Reactive pattern	What expands	Histology clue	Typical association
Follicular hyperplasia	B-cell follicles/ germinal centers	Numerous variable-size secondary follicles; germinal centers with polarity and tingible-body macrophages; architecture generally preserved.	Antigen-driven humoral responses, autoimmune/ infectious stimulation.
Paracortical hyperplasia	T-cell zones	Expanded paracortex, immunoblasts may be prominent, follicles can be compressed.	Viral infections, drug/ immune stimulation.
Sinus histiocytosis	Subcapsular/ medullary sinuses	Dilated sinuses filled with macrophages/histiocytes.	Nodes draining inflammation or tumors.
Suppurative lymphadenitis	Neutrophilic inflammatory foci	Neutrophils, necrosis, microabscess/abscess formation.	Pyogenic bacterial infection.
Granulomatous lymphadenitis	Macrophage aggregates/ granulomas	Epithelioid histiocytes +/- giant cells; may be necrotizing depending cause.	Mycobacteria, fungi, sarcoid-pattern disease, cat-scratch variants, etc.

REACTIVE VS LYMPHOMA: FIRST-PASS ARCHITECTURE RULE

Reactive: usually polymorphic cells, preserved overall architecture, follicles vary in size/shape, germinal centers can show polarity and tingible-body macrophages.

Neoplastic concern: monotony, architectural effacement, repetitive uniform follicles/ sheets, loss of normal compartment relationships, clonal phenotype.

Ozzy checkpoint - click him once you can explain “reactive preserves a pattern; lymphoma tends to replace the pattern.”

DO NOT OVERCALL A BIG GERMINAL CENTER AS LYMPHOMA

Reactive germinal centers can be very large and mitotically active. Look for the **whole pattern**: preserved mantle zones, variation from follicle to follicle, mixed cell populations, tingible-body macrophages, and preserved nodal compartments.

SAL IMAGE SLOT: reactive follicular hyperplasia • paracortical hyperplasia • sinus histiocytosis • suppurative/granulomatous examples actually shown in class

Fast Organ-ID Algorithm

Question	Answer
Deep squamous crypts?	Palatine tonsil.
Ileal wall with aggregated follicles?	Peyer's patch.
Central arteriole inside lymphoid tissue + red pulp?	Spleen.
Lobules + dark cortex/ pale medulla + Hassall corpuscle?	Thymus.
Capsule + subcapsular sinus + cortical follicles + paracortex + medulla?	Lymph node.

THEN ASK THE PATHOLOGY QUESTION

Is normal architecture preserved? Which compartment is expanded? Are the cells polymorphic/reactive or monotonous/effacing?

WHAT CHANGES AFTER SAL

We will keep the normal-organ framework, then swap in the actual class images and adjust the reactive-pattern list to the exact lesions and terminology used by the instructor.

Last 60 Seconds

ORGAN FINGERPRINTS

Node: cortex B / paracortex T / medulla + subcapsular sinus.

Palatine tonsil: squamous crypts.

Peyer's: ileum + aggregated follicles/M cells.

Spleen: red + white pulp, PALS around central arteriole.

Thymus: dark cortex, pale medulla, Hassall corpuscles, no follicles.

Reactive node: architecture generally preserved; identify which compartment is expanded.